

# Solution Synopsis / Executive Summary

ASTRA: Assessment Security, Telemetry & Resilience Architecture

## Failures pause. Exams don't.

With ASTRA, a power cut or a server crash no longer cancels an exam. The exam pauses, candidates continue from exactly where they stopped, and no answer or minute is lost.

### The problem: computer-based exams break down over ordinary faults

**489**

exam centres lost a CUET-UG shift in one day (2022) because the question paper reached them late

**2 days**

in a row, the same Jaipur centre failed: NEET-PG, then an AIIMS exam (Aug 2026)

**~3 hrs**

MPESB candidates waited before a shift was cancelled because a server failed (Jun 2026)

**28 lakh**

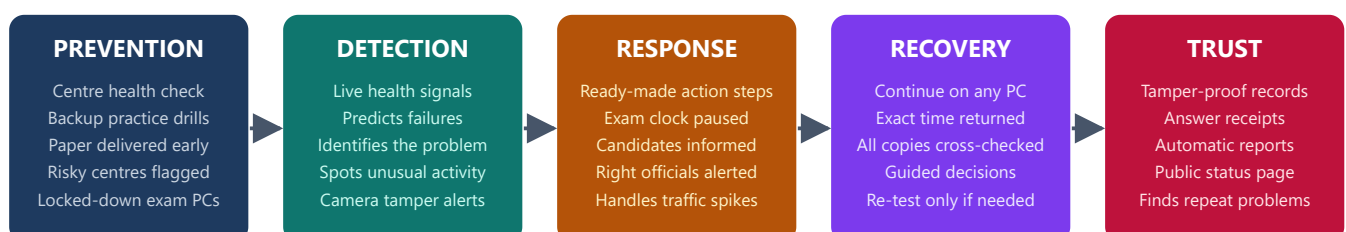
SSC CGL 2025 candidates affected by server crashes and cancelled centres

**Madhya Pradesh** MP has seen all three kinds of digital failure at MPESB:

- **Security failure (2020–21):** three exams were cancelled after question papers were downloaded from unauthorised computers.
- **Technical postponement (May 2025):** a Group-1 exam shift was postponed.
- **Server failure (June 2026):** the Forest Guard / Jail Prahari exam shift was cancelled.

**Why this keeps happening:** nobody sees problems early, there is no automatic backup when something fails, and no system can restore a candidate's exam exactly as it was. So **cancelling the exam becomes the only option.**

### Our solution



#### ★ Answers saved every second: the "Capsule"

A tiny, locked backup of each candidate's exam: answers, question order and time left. It's saved on the exam PC and copied to the centre's own server within 1 second. If a PC or the power fails, the candidate **continues on any PC with exactly the time they had left.**

#### ★ Centres that work without the internet

Each centre runs the exam on its own local network, with a main and a backup server inside the centre. **If the internet or the vendor's central server goes down, the exam carries on.**

### Warns before things break

AI watches live health signals from every centre (power, network, computers) and predicts failures minutes in advance. It also flags risky centres before exam day.

### One dashboard for every level

Invigilators see their room, centre heads see their centre, and the exam authority sees every centre, down to a single seat. CCTV alerts if a camera is covered, moved or replaying old footage.

### Fair, clear decisions

Lost time is measured and returned to each candidate exactly. A step-by-step guide suggests whether to continue, extend, move or re-test, so only affected candidates are re-tested.

### Protected from inside misuse

Even the centre's own server can't read or change answers. Sensitive actions need two people. Every action leaves a record that can't be secretly edited.

## What changes

### Today

- Power cut or server crash → shift cancelled
- Hours of waiting and rumours
- Re-test days or weeks later
- Evidence collected only afterwards

### With ASTRA

- Pause, then continue: **no answers or minutes lost**
- Official update within **30 seconds**
- Only affected candidates re-tested
- A tamper-proof record from the first second

Free, open-source software

Works with any exam vendor

Runs on ordinary centre computers

Rolled out step by step

Hindi + English

Incident figures are from published news reports (sources in the Problem Statement document). ASTRA figures are design goals.